Problem 8

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Problem 1 *Suppose that a stone is thrown vertically upward from a cliff on Mars with an initial velocity of 24 ft/s from a height of 192 ft. The height s of the stone above the ground after t seconds is given by $s(t) = -6t^2 + 24t + 192$.*

(a) *Determine the velocity and acceleration of the stone after t seconds.*

(b) *What is the greatest height of the stone and when does it occur? What are the velocity and acceleration at that time?*

(c) *When does the stone hit the ground? What are the velocity and acceleration at that time?*